CSC4197

RESEARCH METHODOLOGY

ANATOMY OF A RESEARCH PAPER & TYPES OF PAPERS



2 THE IMRAD SKELETON

- Most empirical papers follow **IMRaD**:
- **I**ntroduction — why this problem, why now, what we did
- **M**ethodology — how, in reproducible detail
- **R**esults — what we found (facts, no spin)
- **a**nd
- **D**iscussion — what it means, limitations, implications
- CS adds: **Related Work** (after intro) and often **Conclusion & Future Work**.

3 FULL ANATOMY OF A CS PAPER

1. **Title** — specific, searchable, honest
2. **Authors & affiliations** (order matters — Week 10 ethics)
3. **Abstract** — the paper in 150–250 words
4. **Keywords** — for indexing
5. **Introduction** — context → gap → contribution → roadmap
6. **Related Work** — synthesis + positioning, not a list
7. **Methodology** — design, data, instruments, analysis
8. **Results** — tables, figures, statistics
9. **Discussion** — interpretation, **threats to validity**
10. **Conclusion & Future Work**
11. **Acknowledgments** → **References** → **Appendices**

4 WHAT EACH SECTION MUST ACCOMPLISH

Section	Reader's question it answers
Abstract	Should I read this paper?
Introduction	Why should I care? What's new?
Related Work	Do the authors know the field? Where do they fit?
Methodology	Could I replicate this? Is it sound?
Results	What did the data show?
Discussion	So what? What are the caveats?
Conclusion	What's the one-sentence takeaway?

Reviewers read in roughly this order: abstract → figures → conclusion → methods.

5 THEORY: THE ABSTRACT FORMULA

- Five moves, one sentence-ish each (**C-G-M-R-I**):
  - **Context** — the domain and why it matters
  - **Gap** — what's missing/unknown
  - **Method** — what you did (design, data, scale)
  - **Results** — headline numbers ("improved F1 by 7 points over mBERT")
  - **Implication** — why it matters / who benefits
- **Anti-patterns:** citations in abstract, undefined acronyms, "results will be discussed" (state them!), promises without numbers.

6 CRAFTING TITLES & INTRODUCTIONS

- **Titles:**
  - ✅ "Calibration-Preserving Mortality Prediction for Imbalanced ICU Data"
  - ❌ "A Study on Machine Learning" (vague) ❌ clickbait without substance
- **Introduction funnel (4–6 paragraphs):**
  - broad context → specific problem → gap (with citations) → **"In this paper, we..."** (contributions, often bulleted) → paper roadmap
- **Contribution bullets** should be claims a reviewer can verify: "We release the first public dataset of X (n=...)".

7 TYPES OF RESEARCH PAPERS IN CS

Type	Length	Purpose
Full/regular paper	8–14 pp	Complete study, main venue track
Short paper	4–6 pp	Solid but smaller contribution
Survey/review	20+ pp	Synthesize a field (SLR if systematic)
Tool/demo paper	2–4 pp	Present a working artifact
Dataset paper	varies	Document a reusable dataset
Position paper	4–8 pp	Argue a viewpoint, provoke discussion
Poster/extended abstract	1–2 pp	Early-stage work, networking
Workshop paper	4–8 pp	Emerging ideas, friendlier review

8 CONFERENCE VS. JOURNAL CULTURE IN CS

- **CS is conference-centric** (unusual among disciplines): top conferences (CORE A*/A) are prestige venues with hard deadlines, 15–25% acceptance
- **Journals**: no deadline, multiple revision rounds, often extended versions of conference papers (+30% new material rule)
- Typical pipeline: workshop → conference → journal extension
- Your mini-paper target format: **IEEE conference, 4–6 pages, two columns**

9 CASE STUDY: REVERSE-ENGINEERING A PUBLISHED PAPER

- **In groups:** Using the following top-venue paper (chose any one):
  1. https://aclanthology.org/2025.emnlp-main.154.pdf
  2. https://aclanthology.org/2025.emnlp-main.486.pdf
- Highlight the **C-G-M-R-I** moves in the abstract
- Find the explicit contribution list — verify each is actually delivered in the body
- Locate the gap sentence in Related Work ("However, ...")
- Count figures/tables — what does each *argue*?
- Find the Threats to Validity / Limitations content
- **Debrief:** what structural tricks will you copy for your own paper?

10 FIGURES & TABLES: THE PAPER'S BACKBONE

- Reviewers skim figures **before** reading prose — each must be self-contained (caption tells the full story)
- Tables for exact values; plots for trends/comparisons
- Always: axis labels with units, legible fonts, colorblind-safe palettes, vector graphics (PDF/SVG)
- ❌ Screenshots of Excel charts ❌ 6-pt fonts ❌ "as shown below" (use "Figure 3")

11 HANDS-ON WORKSHOP

- **Abstract bootcamp:**
- (15 min) Each group writes a **C-G-M-R-I abstract** (≤200 words) for your semester project — yes, before results exist: write the *intended* abstract; it forces clarity (it will evolve)
- (10 min) Exchange with another group → mark the 5 moves; flag any missing move
- (5 min) Volunteers read theirs aloud — class identifies moves in real time
- **Due Week 9:** Introduction + Related Work draft (1.5–2 pages, with ≥8 citations) for your project

12 WRAP-UP & NEXT WEEK

- **Key takeaways**
- IMRaD + Related Work = the CS paper skeleton; each section answers a reader question
- Abstract = Context-Gap-Method-Results-Implication
- Know your paper type & venue culture before you write
- **Next week:** Writing tools — LaTeX (Overleaf) vs. MS Word, hands-on lab
- **Bring laptops** · create a free Overleaf or prism account beforehand